Daymark

The morning edition for perishable retail. Every number traceable, every claim replayable, every decision signed — with Gemma running on AMD Instinct.

−58%
waste units vs the spreadsheet rule, 10 weeks of real demand

+27%
forecast skill vs same-weekday baseline, out of sample

<$0.01
inference cost per brief on measured AMD throughput

THE PROBLEM

Fresh food dies on a schedule — and the decision window is 30 minutes

— Every store, every morning: how much to order, what to mark down, what to promote — from memory and spreadsheets.

— Two leaks: **shrink** (thrown-away fresh stock) and **lost sales** (empty shelves on the day demand peaks).

— The EU wastes **153M tonnes** of food a year; the UN target is to halve retail waste by 2030.

— Enterprise AI category management exists — for big chains, at big-chain prices. Mid-size chains have nothing.

WHO WE SERVE

The category manager of a mid-size grocery chain. Not a data scientist. Decides before 7am.

SOURCES: FEEDBACK EU; UN SDG 12.3; INDUSTRY SHRINK RANGES 3–6% OF FRESH SALES

A brief, not a dashboard

— **Orders** for tomorrow — newsvendor targets from quantile forecasts, with the reasons attached.

— **Markdowns** before expiry — costed, with recovered value per line.

— **Promotions** with measured lift from the store's own history, plus ready-to-print creative.

— **Watchlist** where the model is least certain — uncertainty made visible.

— One click to approve, edit, or flag. Every decision signed into a tamper-evident ledger.

DESIGN PRINCIPLE

If it can't be checked, it isn't written.

Facts come from the forecaster; words come from the model; numbers are verified before display.

Same demand. Three policies. Exact arithmetic.

POLICY · STORE 2, 70 DAYS	WASTE UNITS	LOST UNITS	SERVICE	WASTE VALUE
Spreadsheet (last weekday × 1.05)	5,684	26,509	87.8%	$24,190
Forecast-only	3,131	4,494	97.9%	$10,902
Daymark (forecasts + markdowns)	**2,374**	5,842	97.4%	**$8,515**

−58%
waste vs spreadsheet · +9.6 service pts · +24,982 units sold

−24%
the markdown engine alone — with its cost shown openly

Reproduce exactly:
`curl /api/replay?store=2`

WHY AMD IS LOAD-BEARING

Open models on AMD make per-store AI economics real

SERVING STACK

— Gemma served by vLLM on ROCm

— AMD Instinct · AMD Developer Cloud

— OpenAI-compatible endpoint

MEASURED, NOT CLAIMED

— TTFT p50/p95, decode tok/s at concurrency 1/4/8

— $/1M tokens derived from the GPU price

— Per-brief cost shown in the product

WHY IT MATTERS

— Retail data never leaves the retailer

— A brief costs a fraction of a cent

— Feasible per store, not just per chain

Language layer
Gemma · vLLM

Decision core
Rust · deterministic

Forecaster
LightGBM quantiles

BENCHMARK HARNESS: ENGINE/DAYMARK/BENCH.PY → ARTIFACTS/BENCHMARK.JSON

Four checks judges (and retailers) can run

— **Number grounding.** Every figure in the copy is verified against the facts payload; failures regenerate once, then fall back to a labelled template. 17/17 figures checked in the demo brief.

— **Out-of-sample forecasts.** WAPE 26.7% · skill +27.1% · band coverage 78.8% · bias −1.5% on 54,872 held-out item-days.

— **Tamper-evident ledger.** Decisions are SHA-256 chained; the API re-verifies on every read and tests prove tampering is detected.

— **Honest economics.** Revenue is not margin: markdowns trade a little revenue for lower waste — the replay shows the trade-off, not just the win.

From one store to a category

— **Wedge:** fresh-section decisions for a single store — the highest-shrink, highest-uncertainty category.

— **Expansion:** POS/ERP connectors → multi-store roll-up → supplier orders → delivery-note automation (Evolus working prototype in the repo).

— **Timing:** EU food-waste regulation, margin pressure, and open-model inference economics crossing the per-store threshold.

— **Model:** per-store/month SaaS; inference cost visible and tiny; self-hosting keeps retailer data home.

DISTRIBUTION

— Regional grocery associations

— Retail-ops communities

— Consultancies (Vibe / PwC world)

— One pilot store → the chain

CLOSE

The model decides.
The LLM explains.
The ledger proves.

WORKING DEMO

github.com/PhiBao/daymark

Rust core · Next.js edition · Gemma on AMD Instinct · MIT

REINVENT COMMERCE | AMD DEVELOPER CLOUD | ROCM